

# 402Pilot: An x402 Decision Layer for Autonomous Agent Micropayments

Yin Li<sup>1</sup>, Yanbo He<sup>1</sup>, Boo-Ho Yang<sup>2</sup>, Rav Lawana<sup>3</sup>,  
Ziyue Li<sup>4</sup>, Wei Zeng<sup>1</sup>, Jing Tang<sup>1</sup>, Fugee Tsung<sup>1,5\*</sup>

<sup>1</sup>The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China

<sup>2</sup>MOVENSY Inc., Seongnam-si, Republic of Korea

<sup>3</sup>Schneider Electric, Shanghai, China

<sup>4</sup>Technical University of Munich, Munich, Germany

<sup>5</sup>The Hong Kong University of Science and Technology, Hong Kong, China

yligt@connect.hkust-gz.edu.cn, yhe376@connect.hkust-gz.edu.cn, byang@movensys.com,  
rav.lawana@se.com, ziyue.li@tum.de, weizeng@hkust-gz.edu.cn, jingtang@hkust-gz.edu.cn,  
season@ust.hk

## Abstract

Programmable-payment protocols such as x402 enable per-request micropayments, but they do not determine which payable service an autonomous agent should buy under a finite wallet. We formulate this buyer-side problem as *agent-native payment decision-making*: contextual provider selection under wallet pressure, chosen-only paid feedback, and changing market conditions. We propose 402Pilot, a protocol-agnostic buyer-side decision layer between autonomous agents and payment execution that implements purchasing policies for selecting among payable providers. We instantiate it with PA-DCT, a payment-aware discounted contextual Thompson-sampling policy that adapts purchasing decisions under wallet pressure while learning from post-payment feedback. To evaluate buyer-side payment policies, we introduce 402Pilot-Bench, a frozen-replay benchmark spanning 823 tasks, five heterogeneous provider pipelines, and three market regimes, each evaluated over 30 paired seeds. PA-DCT achieves the strongest fixed-wallet adaptive trade-off among non-oracle policies: it maintains competitive service quality while spending only 39–43% of the wallet and reallocates spending as market conditions change. It attains the best non-oracle PA-gap/ $T$  under the price shock and the best mean and worst-case ranks across the nine scenario–metric combinations of quality, ROI, and PA-gap/ $T$ . Comparisons with learning baselines and component ablations further support the effectiveness and design of the proposed decision policy. These results suggest that programmable payment must be complemented by buyer-side decision-making capable of learning service value and adapting purchasing decisions accordingly.

**Code and data** — <https://github.com/MCCodeAI/402Pilot>

## 1 Introduction

The rapid development of x402 brings protocol capabilities that traditional payment flows do not support: machine-readable payment requirements, per-request micropayments, and low-friction settlement. With further advances such as x402 V2 (Reppel, Roscoe, and Nickerson 2025), additional capabilities—service discovery, dynamic service metadata, and pricing—are exposed to agents through programmatic

interfaces, making buyer-side service selection an immediate practical problem.

Several lines of prior work supply pieces of this problem, but not the full buyer-side setting. Budgeted, contextual, and non-stationary bandits offer tools for learning under resource constraints, task-dependent conditions, and changing environments (Badanidiyuru, Kleinberg, and Slivkins 2018; Agrawal and Goyal 2013; Qi, Wang, and Zhu 2023). Cost-aware LLM routing similarly considers quality–cost trade-offs, but typically does not model repeated purchasing decisions under a persistent wallet and post-payment feedback (Chen, Zaharia, and Zou 2024; Ong et al. 2025). These directions do not jointly capture a protocol-mediated paid-feedback loop in which the buyer selects and pays for one service, then learns only from the outcome and cost of that selection. Payment protocols address execution, while auction mechanisms study strategic provider bidding (Patra et al. 2026); neither provides an online spending policy for a single autonomous buyer. Existing work therefore does not jointly address the buyer-side requirements of agent-operated micropayments.

We refer to this buyer-side setting as *agent-native payment decision-making*: an autonomous agent repeatedly makes irreversible purchasing decisions among payable providers under task context and wallet pressure, learns from the performance and cost of each selected service, and adapts to market changes. Unlike human-operated selection, this process must run continuously without human resetting, retuning, or comparison of unchosen outcomes. Micropayments also make repeated low-cost purchases possible, allowing the agent to accumulate feedback over many fine-grained decisions. These properties make autonomous micropayment an online economic decision problem, not only a settlement problem.

We introduce 402Pilot as a protocol-agnostic buyer-side decision layer between the agent and the payment stack. For each request, it derives the task context and obtains the affordable provider set, then applies a buyer policy to select a provider. 402Pilot forwards the selection for payment execution and converts the resulting service outcome and receipt into learning feedback.

\*Corresponding author.

We instantiate this policy as PA-DCT (Payment-Aware Discounted Contextual Thompson Sampling), a contextual bandit that maintains discounted posteriors over service utility and realized cost. It uses Thompson sampling for provider selection under wallet pressure and discounts older observations to adapt as market conditions change.

We evaluate buyer-side policies on *402Pilot-Bench*, a reproducible frozen-replay benchmark with 823 tasks, five heterogeneous provider pipelines, and stationary, reliability-shock, and price-shock scenarios, each evaluated over 30 paired seeds. Among the non-oracle policies, PA-DCT delivers the strongest fixed-wallet adaptive trade-off: it maintains competitive quality while using only 39–43% of the available wallet across all three scenarios and reallocates spending appropriately as reliability and prices change. Against both non-learning policies and learning baselines, it attains the best mean rank (3.3/10) and best worst-case rank (6/10) across the nine scenario–metric comparisons of quality, ROI, and PA-gap/ $T$ . Ablations further clarify how PA-DCT’s components contribute to solvency, adaptation, and robustness.

We make three contributions.

**C1.** We formulate *agent-native payment decision-making*, a new buyer-side online decision problem for autonomous micropayment agents, and introduce *402Pilot* as the missing decision layer above payment execution (§3–§4).

**C2.** We instantiate *402Pilot* with PA-DCT, a payment-aware discounted contextual Thompson-sampling policy that jointly handles wallet pressure, task context, and reliability and price drift, yielding adaptive provider selection under market change (§5).

**C3.** We introduce *402Pilot-Bench* for reproducible evaluation of buyer-side payment policies used by autonomous agents. Its frozen-replay design supports comparisons under paid feedback, finite budgets, and market shocks (§6).

## 2 Related Work

**Programmable payment protocols and discovery.** Programmable payment protocols expose the execution substrate for agent payments. x402 embeds per-request payment into HTTP, and its V2 release adds discovery and service metadata for payable APIs (Reppel, Roscoe, and Nickerson 2025). AP2 provides a payment-agnostic authorization framework for agent-led commerce (Parikh and Surapaneni 2025); A2A–x402 brings x402-style cryptocurrency payments into the Agent-to-Agent protocol (Google Agent Commerce 2025); and A402 binds cryptocurrency payment to service execution through atomic service channels (Li et al. 2026). A recent SoK organizes blockchain-based agent payments around discovery, authorization, execution, and accounting (Zhang et al. 2026). These systems make payable services discoverable and executable, but leave the buyer’s spending policy outside the protocol. *402Pilot* takes that policy as its object of study.

**Budgeted, contextual, and non-stationary bandits.** Classical bandit methods provide relevant algorithmic tools for this setting. Budgeted Thompson Sampling samples reward and cost posteriors under a finite budget (Xia et al. 2015), while bandits with knapsacks give a broader

constrained-resource formulation (Badanidiyuru, Kleinberg, and Slivkins 2018). Contextual bandits estimate action value from task features, with Contextual Thompson Sampling and LinUCB as canonical examples (Agrawal and Goyal 2013; Li et al. 2010). Contextual bandits with knapsacks combine context with resource constraints (Agrawal and Devanur 2016). For non-stationary environments, discounted and sliding-window methods discount or window old observations (Garivier and Moulines 2011; Trovò et al. 2020; Qi, Wang, and Zhu 2023), and weighted linear bandits extend this idea to drifting contextual rewards (Russac, Vernade, and Cappé 2019). However, none of these methods combines budget awareness, task context, and drift adaptation within a protocol-mediated paid-feedback loop.

**Cost-aware LLM routing and provider selection.** FrugalGPT uses cascades to reduce inference cost while preserving quality, while RouteLLM, Hybrid LLM, AutoMix, MixLLM, and LLM Bandit learn prompt-to-model assignments from quality, preference, cost, or latency signals (Chen, Zaharia, and Zou 2024; Ong et al. 2025; Ding et al. 2024; Aggarwal et al. 2024; Wang et al. 2025; Li 2025). These methods typically operate within a fixed serving stack with known candidates and listed prices. *402Pilot* instead studies wallet-constrained endpoint selection with chosen-only feedback on realized outcomes and costs. Reverse-auction work considers strategic provider bidding (Patra et al. 2026), whereas *402Pilot* assumes posted prices and focuses on a single buyer’s online spending policy.

## 3 Buyer-Side Decision Problem

We formalize buyer-side payment decision-making as a sequential provider-selection problem in which an autonomous buyer must allocate a finite wallet across competing services.

### 3.1 Decision Setting

A single autonomous buyer interacts with  $K$  payable providers over a horizon of  $T$  rounds, starting with wallet balance  $B_0 = B$ . At round  $t \in \{0, \dots, T-1\}$ , a task arrives with context  $\mathbf{x}_t \in \mathcal{X}$ , while  $B_t$  denotes the buyer’s current wallet balance.

Each provider  $a \in \{1, \dots, K\}$  carries a charge  $p_{a,t}$ . The buyer may select only from the affordable set

$$\mathcal{A}_t = \{a \in \{1, \dots, K\} : p_{a,t} \leq B_t\}. \quad (1)$$

If  $\mathcal{A}_t = \emptyset$ , the interaction terminates. Let  $\tau \leq T$  denote the resulting number of completed purchases.

Let  $H_t$  denote the information available before selection, including the current task context, wallet state, affordable set, any provider metadata exposed before payment, and feedback from previous purchases. An admissible policy selects only affordable providers:

$$a_t \sim \pi_t(\cdot \mid H_t), \quad \text{supp}(\pi_t(\cdot \mid H_t)) \subseteq \mathcal{A}_t. \quad (2)$$

Let  $\Pi_{\text{adm}}$  denote the class of admissible policies.

The selected provider produces

$$(q_t, f_t) \sim \mathcal{D}_{a_t, t}(\cdot \mid \mathbf{x}_t), \quad (3)$$

where  $q_t \in [0, 1]$  is the realized task quality and  $f_t \in \{0, 1\}$  indicates paid non-delivery: the service is charged but returns no usable output. After payment, the buyer observes  $(q_t, f_t)$  and the realized charge  $c_t = p_{a_t, t}$ , and the wallet evolves as

$$B_{t+1} = B_t - c_t. \quad (4)$$

Outcomes and realized costs are observed only for the selected provider, yielding *chosen-only paid feedback*. Provider outcome distributions and charges may change over time without announced change points.

### 3.2 Buyer Objective

The realized service utility is

$$u_t = q_t - \nu f_t, \quad (5)$$

where  $\nu \geq 0$  penalizes paid non-delivery. The realized charge is normalized as

$$\tilde{c}_t = \text{clip}\left(\frac{c_t}{c_{\max}}, 0, 1\right), \quad c_{\max} > 0, \quad (6)$$

where  $c_{\max}$  is a fixed normalization constant.

Let  $\lambda_t > 0$  denote wallet pressure, computed from the current balance relative to a spending plan. It rises when cumulative spending runs ahead of plan and falls when spending runs behind. Its normalized value is

$$\lambda_{\text{norm}, t} = \frac{\lambda_t}{1 + \lambda_t} \in (0, 1). \quad (7)$$

The payment-aware reward for a completed purchase is

$$r_t = (1 - \lambda_{\text{norm}, t}) u_t - \lambda_{\text{norm}, t} \tilde{c}_t. \quad (8)$$

Thus, increasing wallet pressure shifts weight from service utility to payment.

The buyer seeks an admissible policy maximizing expected cumulative payment-aware reward over the served rounds:

$$\pi^* \in \arg \max_{\pi \in \Pi_{\text{adm}}} \mathbb{E}^{\pi} \left[ \sum_{t=0}^{T-1} r_t \right]. \quad (9)$$

This formulation is independent of any particular decision or learning algorithm.

## 4 The 402Pilot Decision Layer

To operationalize this decision process, we introduce 402Pilot, a policy-agnostic layer between agent buyers and the payment stack. It implements provider selection and post-payment learning through interchangeable buyer policies. Figure 1 shows the per-round decision flow.

Upstream components provide a task request and candidate payment requirements

$$\mathcal{Q}_t = \{(a, p_{a,t})\}_{a=1}^K.$$

The context encoder maps the request to  $\mathbf{x}_t$ , while the wallet uses  $\mathcal{Q}_t$  to derive the affordable set  $\mathcal{A}_t$  and wallet pressure  $\lambda_t$ . Wallet pressure compares cumulative spending with the expected budget trajectory:

$$\lambda_t = \begin{cases} \lambda_0, & t = 0, \\ \lambda_0 \exp \left[ \alpha \left( \frac{(B - B_t)/B}{t/T} - 1 \right) \right], & t > 0, \end{cases} \quad (10)$$

where  $\lambda_0 > 0$  is the baseline pressure and  $\alpha \geq 0$  controls its sensitivity to deviations from the spending plan. The exponential form keeps pressure positive and smoothly increases cost sensitivity as overspending grows. The policy then invokes  $\text{select}(\mathbf{x}_t, \mathcal{A}_t) \rightarrow a_t \in \mathcal{A}_t$  and passes the selected provider to the underlying payment substrate, such as x402, for execution and settlement. After the transaction, the outcome scorer converts the provider response and payment receipt into the realized utility  $u_t$  and charge  $c_t$ . The wallet records  $c_t$ , and the policy receives the post-payment feedback through  $\text{update}(\mathbf{x}_t, a_t, u_t, c_t)$ .

The interface restricts selection to  $\mathcal{A}_t$  and exposes feedback only for the chosen provider, consistent with the affordability and chosen-only feedback setting in Section 3. It supports diverse buyer policies, including heuristic, optimization-based, and learning-based approaches.

## 5 The PA-DCT Policy

As a concrete buyer policy within 402Pilot, we introduce Payment-Aware Discounted Contextual Thompson Sampling (PA-DCT), a contextual bandit. PA-DCT operates as a continual decision loop: it maintains discounted Bayesian posteriors over provider utility and cost, draws Thompson samples, and uses wallet pressure to select among affordable providers. After each transaction, realized utility and cost update the posteriors, allowing the policy to adapt as provider performance and prices change.

### 5.1 Discounted Utility and Cost Posteriors

PA-DCT conditions its estimates on task context through a discrete bucket map  $k_t = k(\mathbf{x}_t) \in \{1, \dots, K_b\}$ . For each provider-context pair  $(a, k)$ , it maintains two discounted Bayesian posteriors: a Q-posterior over realized utility  $u_t = q_t - \nu f_t$  and a C-posterior over realized cost  $c_t$ .

Each cell stores a shared effective count  $\bar{n}_t^{a,k}$ , a utility sum  $\bar{S}_t^{a,k}$ , and a cost sum  $\bar{S}_{c,t}^{a,k}$ . Each completed transaction contributes one observation to both posteriors. To remain responsive to changing reliability and prices, PA-DCT applies a common discount factor  $\gamma \in (0, 1]$  to every cell at the start of each round. The selection-time statistics are

$$\begin{aligned} \bar{n}_t^{a,k} &= \gamma \bar{n}_{t-1}^{a,k}, \\ \bar{S}_t^{a,k} &= \gamma \bar{S}_{t-1}^{a,k}, \\ \bar{S}_{c,t}^{a,k} &= \gamma \bar{S}_{c,t-1}^{a,k}. \end{aligned} \quad (11)$$

The Q-posterior uses prior mean  $\mu_{0,q}$ , prior variance  $\sigma_{0,q}^2$ , and likelihood variance  $\sigma_q^2$ . Its posterior variance and mean are

$$\begin{aligned} \Sigma_q^{a,k} &= \left( \frac{1}{\sigma_{0,q}^2} + \frac{\bar{n}_t^{a,k}}{\sigma_q^2} \right)^{-1}, \\ \theta_q^{a,k} &= \Sigma_q^{a,k} \left( \frac{\mu_{0,q}}{\sigma_{0,q}^2} + \frac{\bar{S}_t^{a,k}}{\sigma_q^2} \right). \end{aligned} \quad (12)$$

where  $\theta_q^{a,k}$  and  $\Sigma_q^{a,k}$  denote the posterior mean and variance of the utility model, respectively.

The C-posterior follows the same Normal-Normal construction. Its posterior mean and variance  $(\theta_c^{a,k}, \Sigma_c^{a,k})$  are

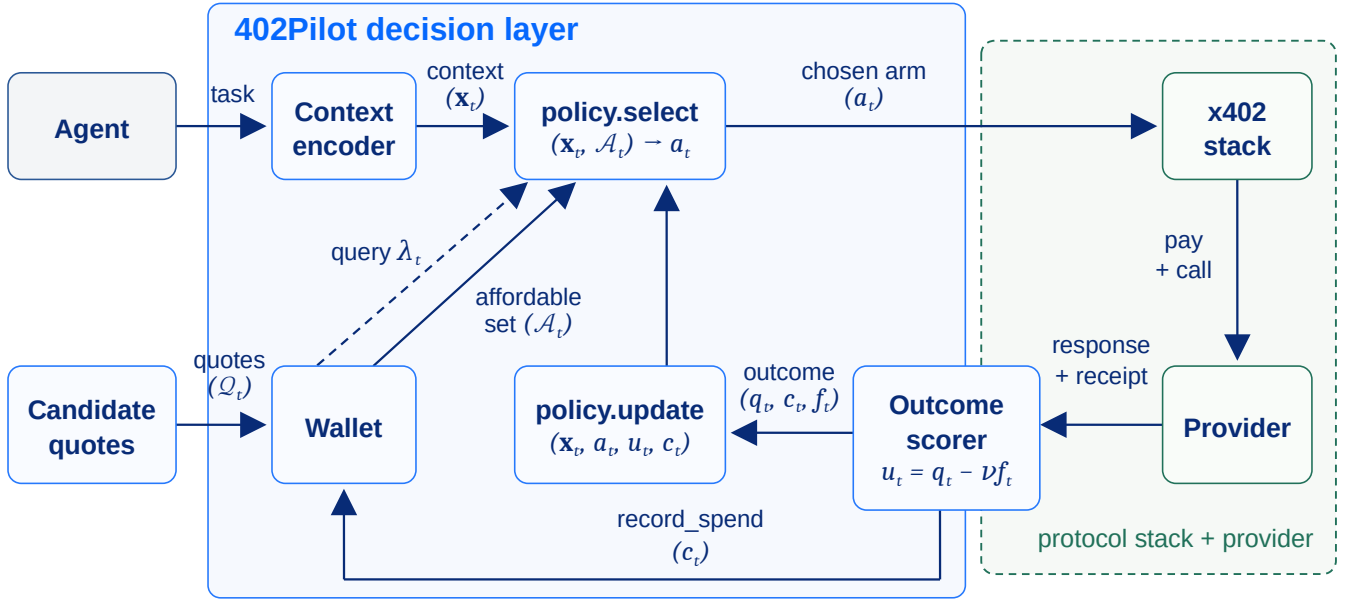


Figure 1: The 402Pilot per-round decision flow. Upstream components supply a task and payable candidates  $\mathcal{Q}_t$ . Within 402Pilot, the context encoder derives the task context  $\mathbf{x}_t$ , while the wallet separately exposes the affordable set  $\mathcal{A}_t$  and wallet pressure  $\lambda_t$ . The policy selects  $a_t$ ; the payment substrate executes the selected transaction; and the returned response and receipt yield  $(q_t, f_t, c_t)$ . The layer then reports  $c_t$  to the wallet and passes  $(\mathbf{x}_t, a_t, u_t, c_t)$  to the policy update. The figure shows x402 as the payment substrate, but the same boundary can be implemented by another programmable-payment substrate.

computed analogously using prior  $\mathcal{N}(\bar{c}_a, \sigma_{0,c}^2)$ , likelihood variance  $\sigma_c^2$ , and discounted cost sum  $\bar{S}_{c,t}^{a,k}$ . The initial specification cost  $\bar{c}_a$  defines the cost prior, while observed receipt charges continually update the posterior as provider prices change.

All provider–context cells are discounted once per round, including those not selected in the current round, so stale evidence gradually loses influence over time.

## 5.2 Selection Rule

At decision time, PA-DCT implements `select` using the task context  $\mathbf{x}_t$ , the affordable provider set  $\mathcal{A}_t$ , and the wallet pressure  $\lambda_t$ . Write  $\lambda_{\text{norm}} = \lambda_{\text{norm},t}$  for the normalized wallet pressure in Eq. (7). The context bucket determines the corresponding provider–context cells, from which Thompson samples are drawn as

$$\begin{aligned} \hat{u}^{a,k_t} &\sim \mathcal{N}(\theta_q^{a,k_t}, \Sigma_q^{a,k_t}), \\ \hat{c}^{a,k_t} &\sim \mathcal{N}(\theta_c^{a,k_t}, \Sigma_c^{a,k_t}). \end{aligned} \quad (13)$$

The sampled cost is first normalized, after which the sampled utility and normalized cost are combined into the following payment-aware reward:

$$\begin{aligned} \tilde{c}^{a,k_t} &= \text{clip}(\hat{c}^{a,k_t}/c_{\max}, 0, 1), \\ \hat{r}_a &= (1 - \lambda_{\text{norm}})\hat{u}^{a,k_t} - \lambda_{\text{norm}}\tilde{c}^{a,k_t}. \end{aligned} \quad (14)$$

The sampled reward in Eq. (14) is the pre-payment counterpart of the realized reward in Eq. (8): it replaces realized

### Algorithm 1 PA-DCT policy loop.

**Require:** Initial specification costs  $\bar{c}_a$ ; discount  $\gamma$ ; normalizer  $c_{\max}$ .

**Require:** Context encoder  $k(\cdot)$  and Q/C posterior priors.

- 1: **for**  $(a, k) \in \{1, \dots, K\} \times \{1, \dots, K_b\}$  **do**
- 2:    $n^{a,k} \leftarrow 0$ ;    $S^{a,k} \leftarrow 0$ ;    $S_c^{a,k} \leftarrow 0$
- 3: **for** each served round  $t$  with  $\mathcal{A}_t \neq \emptyset$  **do**
- 4:   Receive  $\mathbf{x}_t$  and  $\mathcal{A}_t$
- 5:   For all  $(a, k)$ , multiply  $n^{a,k}$ ,  $S^{a,k}$ , and  $S_c^{a,k}$  by  $\gamma$
- 6:   Read  $\lambda_t$ ;    $k_t \leftarrow k(\mathbf{x}_t)$ ;    $\lambda_{\text{norm}} \leftarrow \lambda_t/(1 + \lambda_t)$
- 7:   **for**  $a \in \mathcal{A}_t$  **do**
- 8:     Compute the Q/C posterior parameters for  $(a, k_t)$
- 9:     Sample  $\hat{u}^{a,k_t}$  and  $\hat{c}^{a,k_t}$  using Eq. (13)
- 10:     $\tilde{c}^{a,k_t} \leftarrow \text{clip}(\hat{c}^{a,k_t}/c_{\max}, 0, 1)$
- 11:     $\hat{r}_a \leftarrow (1 - \lambda_{\text{norm}})\hat{u}^{a,k_t} - \lambda_{\text{norm}}\tilde{c}^{a,k_t}$
- 12:    Select  $a_t \leftarrow \arg \max_{a \in \mathcal{A}_t} \hat{r}_a$
- 13:    Receive realized  $(u_t, c_t)$
- 14:     $n^{a_t,k_t} += 1$ ;    $S^{a_t,k_t} += u_t$ ;    $S_c^{a_t,k_t} += c_t$

utility and normalized cost with Thompson samples. As wallet pressure increases, selection becomes progressively more cost-sensitive while remaining driven by posterior samples rather than fixed cost thresholds. The provider with the highest sampled reward is selected:  $a_t = \arg \max_{a \in \mathcal{A}_t} \hat{r}_a$ .

### 5.3 Learning from Post-Payment Feedback

After the selected transaction completes, PA-DCT implements update using the realized utility  $u_t$  and receipt cost  $c_t$ . The task is mapped to the corresponding context bucket  $k_t$ , and the observations are incorporated only into the selected provider–context cell:

$$\begin{aligned} n_t^{a_t, k_t} &= \bar{n}_t^{a_t, k_t} + 1, \\ S_t^{a_t, k_t} &= \bar{S}_t^{a_t, k_t} + u_t, \\ S_{c_t}^{a_t, k_t} &= \bar{S}_{c_t}^{a_t, k_t} + c_t. \end{aligned} \quad (15)$$

All other provider–context cells retain their discounted statistics from Eq. (11).

Algorithm 1 summarizes the PA-DCT policy loop. Appendix B provides formal support for its key components and discusses the scope of the analysis.

## 6 Evaluation

Evaluating buyer-side payment policies requires a benchmark with diverse tasks, heterogeneous providers, controllable market scenarios, and reproducible execution. Existing LLM benchmarks do not jointly satisfy these requirements, so we construct 402Pilot-Bench by extending public datasets with multiple provider pipelines, prices, and market dynamics. Under its frozen-replay design, multiple responses are pre-generated and scored for each task–provider pair, then sampled from the same frozen pool across policies, enabling reproducible comparisons without live-inference variability. Since frozen replay abstracts payment execution, a local x402 witness separately validates the decision-layer interface against an actual HTTP 402 payment flow; implementation details appear in Appendix G.

### 6.1 Experimental Setup

**Benchmark.** *402Pilot-Bench* contains 823 effective tasks across the four datasets in Table 1 (Chen et al. 2021; Yang et al. 2018; Joshi et al. 2017; Köpf et al. 2023), which define the task buckets used by PA-DCT. Utility scores are normalized to  $[0, 1]$  using the dataset-specific metrics shown in the table. For each task–provider pair, we cache five independently generated responses with precomputed utility scores. Across the five providers described below, this yields  $823 \text{ tasks} \times 5 \text{ providers} \times 5 \text{ responses} = 20,575$  scored responses. All scenarios replay the same task pool and response cache; only realized failures and effective prices change.

**Providers.** Table 2 summarizes the five providers. P-cheap, P-mid, and P-premium represent different price–quality levels. P-adv and P-flaky simulate two problems that buyers may face in an open service market: fraudulent or misrepresented service quality, and unstable service that may fail after payment. Because both have the same listed price as P-mid, their true performance must be learned from observed outcomes.

**Scenario design.** The scenarios test whether a policy can maintain a useful allocation when provider conditions remain stable or change without warning. S1 is stationary. In S2, P-mid experiences a reliability shock, with a 30% forced-failure

| Dataset / task type               | Tasks      | Utility metric         |
|-----------------------------------|------------|------------------------|
| HumanEval / coding                | 164        | Execution-based pass@1 |
| HotpotQA / multi-hop QA           | 220        | Max normalized EM/F1   |
| TriviaQA-web / closed-form web QA | 219        | Max normalized EM/F1   |
| OpenAssistant / open-ended QA     | 220        | Cached LLM-as-judge    |
| <b>Total</b>                      | <b>823</b> | –                      |

Table 1: Composition of 402Pilot-Bench.

| Provider  | Price  | Pipeline                                                          |
|-----------|--------|-------------------------------------------------------------------|
| P-cheap   | 0.0005 | qwen3.5-flash; uniform prompt; <i>cheap tier</i>                  |
| P-mid     | 0.002  | GPT-5.4-mini; uniform prompt; <i>mid tier</i>                     |
| P-premium | 0.01   | GPT-5.4; uniform prompt; <i>premium tier</i>                      |
| P-adv     | 0.002  | GPT-5.4-mini; 60% adversarial prompt; <i>fluent wrong answers</i> |
| P-flaky   | 0.002  | GPT-5.4-mini; 40% timeout injection; <i>billed timeouts</i>       |

Table 2: Provider set used in 402Pilot-Bench. P-mid, P-adv, and P-flaky share the same price tier and base model.

rate from rounds 3,000 to 5,500 before recovery. In S3, P-premium’s price drops from \$0.01 to \$0.002 at round 1,000, testing whether the policy shifts toward a newly attractive provider.

Each policy is evaluated for  $T = 10,000$  rounds with a \$50 budget over 30 paired seeds. This budget is deliberately binding: at the initial premium price, it covers only 5,000 of the 10,000 rounds. For each seed, all policies are evaluated using the same task order, cached-response draws, and scenario changes.

**Hyperparameters.** The discount factor  $\gamma$ , paid-non-delivery penalty  $\nu$ , and cost scale  $c_{\max}$  most directly govern the policy’s adaptation, failure penalty, and cost normalization. The discount factor  $\gamma = 0.999$  provides an effective memory of about  $10^3$  rounds, balancing responsiveness with stable estimation. The paid-non-delivery penalty  $\nu = 0.5$  makes billed failures materially worse without overwhelming the  $[0, 1]$  utility signal. The cost scale  $c_{\max} = \$0.01$  maps the initial premium price to 1 and keeps all lower prices on the same normalized scale. All remaining settings are fixed; complete notation and values appear in Appendix A, with sensitivity analyses in Appendix D.

**Comparators.** The comparators cover non-learning policies, learning policies, and an oracle upper bound. The non-learning group includes Random, three fixed-arm policies, and BudgetRule, capturing uninformed selection, static provider preferences, and a hand-crafted wallet rule. BudgetRule switches from P-premium to P-mid and then P-cheap when the remaining budget falls below 50% and 20%, respectively.

The learning group covers the main capabilities relevant to buyer-side routing. Contextual DS-TS (Qi, Wang, and Zhu

2023) tracks utility drift but lacks wallet pressure; Contextual BTS (Xia et al. 2015) learns utility and cost under a finite budget but does not discount stale observations; PM-Greedy, inspired by FrugalGPT-style routing (Chen, Zaharia, and Zou 2024), routes using listed prices rather than learned realized costs; and LinCBwK-Adapt (Agrawal and Devanur 2016) combines context and budget constraints but lacks explicit drift discounting.

The replay True Oracle selects the affordable arm with the highest realized payment-aware reward and is reported only as an unattainable upper bound. Implementation details and replay adaptations of all comparators appear in Appendix F.

## 6.2 Metrics

Evaluating buyer-side decisions under a fixed wallet requires considering both the service quality delivered and how the budget is used. We therefore report complementary metrics covering delivered quality, wallet consumption, spending efficiency, and performance relative to the replay upper bound. Table 3 gives their calculations.

Full-horizon mean quality measures the service quality delivered over all  $T$  rounds, with rounds after budget exhaustion counted as zero. Budget use captures the share of the initial wallet consumed. ROI measures the quality obtained per dollar spent and is used as an efficiency diagnostic rather than an optimization objective. PA-gap/ $T$  measures the per-round cumulative payment-aware reward gap relative to the replay True Oracle, providing an empirical replay counterpart of the buyer objective in Eq. (9).

| Metric                 | Calculation                                 | Meaning                                       |
|------------------------|---------------------------------------------|-----------------------------------------------|
| $\bar{q}_T \uparrow$   | $(1/T) \sum_{t < \tau} q_t$                 | Mean quality; exhausted rounds count as zero. |
| Budget use             | $100 \sum_{t < \tau} c_t / B$               | Wallet share spent.                           |
| ROI $\uparrow$         | $\sum_{t < \tau} q_t / \sum_{t < \tau} c_t$ | Quality per dollar; efficiency only.          |
| PA-gap/ $T \downarrow$ | $(R_T^{\text{TO}} - R_T^{\pi})/T$           | Gap to replay True Oracle.                    |

Table 3: Evaluation metrics.

## 6.3 Main Results

PA-DCT delivers the strongest fixed-wallet adaptive trade-off on 402Pilot-Bench: it maintains competitive service quality under controlled spend and reallocates in the right direction when reliability or prices change. Across the three scenarios in Table 4, PA-DCT consistently uses only 39–43% of the wallet while remaining among the strongest non-oracle policies on PA-gap/ $T$ , including the best non-oracle result in S3. Together, these results show that the objective of buyer-side payment decision-making is not to maximize service quality in every market, but to learn when additional quality is worth paying for under a finite wallet. Accordingly, PA-DCT is not intended to maximize pointwise quality in every scenario: policies such as LinCBwK-Adapt and premium-heavy rules occasionally achieve higher absolute quality, but only by consuming substantially more of the wallet, resulting in weaker payment-aware trade-offs.

The three scenarios highlight different aspects of PA-DCT’s behavior. In the stationary market (S1), Always-P-mid achieves the best non-oracle PA-gap/ $T$  with similarly low budget use, reflecting the advantage of a fixed policy when its selected tier matches a stable market. PA-DCT pays a modest exploration cost but still uses only 42% of the wallet; by contrast, BudgetRule and LinCBwK-Adapt obtain slightly higher quality while consuming 80% and 93%, respectively. This highlights the role of wallet pressure in preserving long-horizon budget efficiency. In the reliability shock (S2), PA-DCT is not significantly different from the numerically best non-oracle baseline, Always-P-cheap, on PA-gap/ $T$ , while delivering substantially higher service quality (0.761 vs. 0.610) through reallocation away from the failing mid-tier provider. This highlights the role of reliability adaptation in preserving service quality. In the price shock (S3), PA-DCT achieves the best non-oracle PA-gap/ $T$ . Several learning baselines attain slightly higher absolute quality, but their gains come with substantially worse PA-gap/ $T$  (0.195–0.264 vs. 0.121) and lower ROI. This highlights the role of price adaptation in reallocating purchases as provider prices change while keeping spending under control.

Appendix E provides a direct view of these allocation shifts, showing PA-DCT moving away from P-mid during the outage, returning after recovery, and shifting toward discounted P-premium after the price promotion.

A closer look at the three types of baselines shows how their underlying design principles are reflected in the experimental results. Among the non-learning policies, fixed-arm strategies perform well only when their selected tier matches the market, while BudgetRule follows wallet thresholds rather than provider feedback. The Thompson-sampling baselines expose a split between adaptation and budget control. Contextual DS-TS responds to market changes but exhausts the wallet in S1 and S2, whereas Contextual BTS preserves the wallet but remains near the cheap tier and therefore misses the discounted premium opportunity in S3. The cost-aware learning baselines provide the strongest quality comparison, but not the strongest payment-aware trade-off. PM-Greedy and LinCBwK-Adapt achieve competitive quality, but consistently at the cost of substantially higher wallet consumption and worse PA-gap/ $T$ . Taken together, these results show that none of the baselines simultaneously maintains budget discipline and adapts effectively across reliability and price changes. The ablation study below separates the roles of PA-DCT’s components in achieving this balance.

## 6.4 Ablation Study

The ablations further clarify the role of each PA-DCT component. To complement the endpoint metrics, AdaptT measures the number of post-shock rounds required for sustained recovery (lower is better); complete ablation definitions and results appear in Appendix C. Payment-aware ranking primarily preserves full-horizon solvency: removing it ( $-P$ ) exhausts every S1/S2 wallet and increases PA-gap/ $T$  from 0.166 to 0.878 in S2, while full-horizon mean quality falls from 0.761 to 0.517. Adaptation depends on both temporal discounting and realized-cost learning. Removing discounting ( $-D$ ) slows S2 recovery by 53% (1,467  $\rightarrow$  2,249 rounds),

| Policy               | S1 Stationary        |                |                |                        | S2 Mid Outage        |                |                |                        | S3 Premium Promo     |                |                |                        |
|----------------------|----------------------|----------------|----------------|------------------------|----------------------|----------------|----------------|------------------------|----------------------|----------------|----------------|------------------------|
|                      | $\bar{q}_T \uparrow$ | Budget use (%) | ROI $\uparrow$ | PA-gap/ $T \downarrow$ | $\bar{q}_T \uparrow$ | Budget use (%) | ROI $\uparrow$ | PA-gap/ $T \downarrow$ | $\bar{q}_T \uparrow$ | Budget use (%) | ROI $\uparrow$ | PA-gap/ $T \downarrow$ |
| Random               | 0.688                | 66             | 209            | 0.365                  | 0.676                | 66             | 205            | 0.375                  | 0.688                | 37             | 370            | 0.274                  |
| Always-P-cheap       | 0.610                | 10             | <b>1,220</b>   | 0.167                  | 0.610                | 10             | <b>1,220</b>   | <b>0.164</b>           | 0.610                | 10             | <b>1,220</b>   | 0.195                  |
| Always-P-mid         | 0.819                | 40             | 410            | <b>0.101</b>           | 0.757                | 40             | 379            | 0.174                  | 0.819                | 40             | 410            | 0.129                  |
| Always-P-premium     | 0.433                | 100            | 87             | 1.072                  | 0.433                | 100            | 87             | 1.070                  | <b>0.865</b>         | 56             | 309            | 0.400                  |
| BudgetRule           | <b>0.831</b>         | 80             | 208            | 0.692                  | 0.769                | 80             | 192            | 0.722                  | 0.859                | 56             | 307            | 0.405                  |
| Contextual DS-TS     | 0.559                | 100            | 112            | 0.858                  | 0.514                | 100            | 103            | 0.884                  | 0.840                | 48             | 349            | 0.264                  |
| Contextual BTS       | 0.612                | 10             | 1,187          | 0.169                  | 0.612                | 10             | 1,187          | 0.166                  | 0.612                | 10             | 1,187          | 0.197                  |
| PM-Greedy            | 0.732                | 90             | 179            | 0.542                  | 0.627                | 99             | 128            | 0.660                  | 0.836                | 44             | 384            | 0.198                  |
| LinCBwK-Adapt        | 0.824                | 93             | 180            | 0.457                  | <b>0.796</b>         | 100            | 159            | 0.515                  | 0.836                | 45             | 377            | 0.195                  |
| <b>PA-DCT (ours)</b> | 0.797                | 42             | 378            | 0.133                  | 0.761                | 43             | 357            | 0.166                  | 0.831                | 39             | 429            | <b>0.121</b>           |
| True Oracle (UB)     | 0.901                | 32             | 561            | 0.000                  | 0.901                | 33             | 551            | 0.000                  | 0.906                | 25             | 722            | 0.000                  |

Table 4: Main results on 402Pilot-Bench, averaged over 30 paired seeds. Rows are grouped as non-learning baselines, learning baselines, PA-DCT, and the replay upper bound. Bold marks the best non-oracle entry in the quality, ROI, and PA-gap/ $T$  columns.

whereas replacing learned realized costs with specification prices ( $-C_{\text{post}}$ ) largely eliminates price adaptation in S3, reducing post-shock P-premium selection from 66.1% to 3.6% and increasing PA-gap/ $T$  from 0.121 to 0.144.

Context and Thompson sampling mainly improve adaptation efficiency and robustness. Removing contextual buck-eting ( $-C$ ) delays S3 AdaptT from 200 to 398 rounds with little change in endpoint performance. Replacing Thompson sampling with posterior means ( $-TS$ ) increases the S1 PA-gap/ $T$  standard deviation from 0.004 to 0.046 and delays S3 AdaptT from 200 to 2,232 rounds, with only 24/30 seeds reaching the AdaptT threshold. Overall, the ablations assign distinct operational roles to the components: payment awareness preserves solvency, discounting and realized-cost learning enable adaptation, context accelerates recovery, and Thompson sampling improves robustness.

## 7 Discussion and Future Work

Recent advances in programmable payment protocols and community-driven reputation mechanisms provide an increasingly capable infrastructure for agentic commerce. Our results suggest that infrastructure alone is not sufficient for autonomous payment, because agents must still decide how to allocate a limited budget as service value and market conditions evolve. This motivates buyer-side decision-making as a distinct layer between agents and payment protocols. PA-DCT serves as an initial instantiation of this decision layer rather than a definitive solution; future work may explore reinforcement learning, foundation-model-based policies, or other decision mechanisms within the same framework.

Our evaluation emphasizes reproducible policy comparison rather than complete deployment validation. Frozen replay isolates buyer-side decision quality under identical service traces, while the local x402 witness demonstrates end-to-end integration with an HTTP 402 payment loop. Extending evaluation to controlled live-market deployments remains important future work.

The benchmark also covers a finite set of providers, tasks, and market conditions, and learning quality ultimately de-

pends on the reliability of outcome feedback. Future work should expand the benchmark to broader industrial scenarios, richer provider ecosystems, more complex market dynamics, and more reliable outcome evaluation.

## 8 Conclusion

Programmable payment protocols enable agents to execute payments, but they do not determine what is worth paying for. We introduced 402Pilot as a buyer-side decision layer between agents and payment protocols, together with PA-DCT as an initial payment-aware decision policy for budget-constrained service selection. Evaluation on 402Pilot-Bench showed that PA-DCT achieves a strong fixed-wallet trade-off while adapting its purchasing decisions under reliability and price changes. These results establish buyer-side decision-making as a new research direction for agentic commerce and motivate future work on richer decision policies, broader benchmarks, and real-world deployment.

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## A Notation and Locked Settings

This appendix collects the notation and locked experimental settings used throughout the paper and Appendix B.

## B Proofs

This appendix provides formal support for the main analytical properties of PA-DCT rather than a complete analysis of the replay system. Specifically, we establish reward boundedness (Prop. 1), posterior consistency in the stationary no-discount regime (Prop. 2), a stationary well-specified regret reduction (Thm. 1), and a fixed-schedule adaptation rate after an abrupt change (Thm. 2). The regret theorem is intentionally stated for a clean version of the score; §B.7 records the gap to the full empirical setting.

### B.1 Shared assumptions

We use the notation of §3 and §5 of the main paper, summarized in Appendix A. All regret sums below are over served rounds with nonempty affordable set; if a wallet exhausts early, replace  $T$  by the served horizon  $\tau$ . The formal claims use the following assumptions. Different results require different subsets of these assumptions; only the stationary regret theorem relies on the stronger assumptions (A2')–(A3') introduced below.

- (A1) *Stationary or piecewise-stationary cells.* Conditional on the chosen arm  $a$  and bucket  $k$ , and averaging over the exogenous task draw within that bucket, observations are independent across pulls and the joint distribution of  $(q_t, c_t, f_t)$  is identical within each phase.
- (A2) *Sub-Gaussian noise.* The utility  $u_t$  given cell  $(a, k)$  is  $\sigma_u$ -sub-Gaussian around mean  $\mu_u^{a,k}$ ; the realized cost  $c_t$  is  $\sigma_{c,\text{true}}$ -sub-Gaussian around mean  $\mu_c^{a,k}$ . The policy uses fixed proxy variances  $\sigma_q^2, \sigma_c^2$ , possibly different from the true variances.
- (A3) *Bounded means.*  $|\mu_u^{a,k}| \leq 1$  and  $\mu_c^{a,k} \in [0, c_{\max}]$  for every  $(a, k)$ .

The stationary regret reduction further uses two restrictions that are not needed by the boundedness, consistency, or adaptation claims:

- (A2') *Well-specified product model.* The policy's proxy likelihoods are the true likelihoods: utility and cost observations are conditionally independent Gaussians around  $(\mu_u^{a,k}, \mu_c^{a,k})$  with variances  $\sigma_q^2 = \sigma_u^2$  and  $\sigma_c^2 = \sigma_{c,\text{true}}^2$ , and the prior factorizes as the policy's independent Gaussian Q/C priors.
- (A3') *Clean score.* The regret reduction is stated for the unclipped score  $(1 - \lambda_{\text{norm}})\hat{u}^{a,k} - \lambda_{\text{norm}}\hat{c}^{a,k}/c_{\max}$ , or equivalently for trajectories on which the implementation's cost-sample clip in the PA-DCT sampled reward (Eq. (14) of the main paper) is inactive.

The posterior arguments use the Normal–Normal mean identities

$$\theta_q^{a,k} = \frac{\kappa_q \mu_{0,q} + S_t^{a,k}}{\kappa_q + n_t^{a,k}}, \quad \theta_c^{a,k} = \frac{\kappa_c \bar{c}_a + S_{c_t}^{a,k}}{\kappa_c + n_t^{a,k}}, \quad (16)$$

where  $\kappa_q = \sigma_q^2/\sigma_{0,q}^2$  and  $\kappa_c = \sigma_c^2/\sigma_{0,c}^2$ . We write a single  $n_t^{a,k}$  because each selected call supplies one utility and one cost observation. If quality and cost discounts are split, the same formulas apply to the two posteriors with their own effective counts.

### B.2 Proposition 1: reward boundedness

**Proposition 1** (Reward boundedness). *For every round  $t$  with  $0 \leq \nu \leq 1$ , the payment-aware reward satisfies  $r_t \in [-1, +1]$ .*

*Proof.* Since  $q_t \in [0, 1]$ ,  $f_t \in \{0, 1\}$ , and  $0 \leq \nu \leq 1$ , we have  $u_t = q_t$  when  $f_t = 0$  and  $u_t = q_t - \nu \in [-1, 1]$  when  $f_t = 1$ . Thus  $|u_t| \leq 1$ ; also  $\tilde{c}_t \in [0, 1]$  and  $\lambda_{\text{norm}} \in (0, 1)$ . Hence  $|r_t| \leq (1 - \lambda_{\text{norm}})|u_t| + \lambda_{\text{norm}}|\tilde{c}_t| \leq (1 - \lambda_{\text{norm}}) + \lambda_{\text{norm}} = 1$ .  $\square$

### B.3 Proposition 2: posterior consistency

**Proposition 2** (Posterior consistency, stationary regime). *Fix  $(a, k)$  and assume the utility distribution at  $(a, k)$  is stationary with finite variance and mean  $\mu_u^{a,k}$ . With no discount ( $\gamma = 1$ ), the  $Q$ -posterior mean  $\theta_q^{a,k}$  converges almost surely to  $\mu_u^{a,k}$  as the cell's pull count  $n^{a,k} \rightarrow \infty$ . The same holds for the  $C$ -posterior and  $\mu_c^{a,k}$ .*

*Proof.* Let  $\{u_t^{(i)}\}_{i=1}^n$  be the utility observations in cell  $(a, k)$  after  $n$  pulls. By (A1), observations within this stationary cell are i.i.d.; the strong law of large numbers therefore gives  $S_t^{a,k}/n_t^{a,k} = n^{-1} \sum_i u_t^{(i)} \rightarrow \mu_u^{a,k}$  almost surely. Dividing the first expression in Eq. (16) by  $n_t^{a,k}$  in numerator and denominator gives  $\theta_q^{a,k} \rightarrow \mu_u^{a,k}$  because  $\kappa_q/n_t^{a,k} \rightarrow 0$ . The cost posterior is identical with  $\bar{c}_a, S_{c_t}^{a,k}, \kappa_c$  in place of  $\mu_{0,q}, S_t^{a,k}, \kappa_q$ .  $\square$

### B.4 Lemma: discounted effective sample size

**Lemma 1** (Effective sample size). *For any  $\gamma \in (0, 1)$  and any sequence of pulls of cell  $(a, k)$ ,*

$$n_t^{a,k} \leq \frac{1 - \gamma^{t+1}}{1 - \gamma} \leq \frac{1}{1 - \gamma}.$$

*The first bound is tight when  $(a, k)$  is pulled at every round.*

*Proof.* If  $\mathcal{T}_t^{a,k} \subseteq \{0, \dots, t\}$  is the set of pull rounds for cell  $(a, k)$ , the round-based discount rule gives  $n_t^{a,k} = \sum_{s \in \mathcal{T}_t^{a,k}} \gamma^{t-s} \leq \sum_{j=0}^t \gamma^j = (1 - \gamma^{t+1})/(1 - \gamma)$ .  $\square$

### B.5 Stationary analysis

We now turn from posterior properties to the stationary decision rule. Under the well-specified product-Gaussian model and clean-score restriction, PA-DCT induces a two-dimensional linear Gaussian Thompson-sampling problem.

**Theorem 1** (Clean-score regret reduction to Gaussian Thompson sampling). *Under (A1), (A2'), (A3'), and  $\gamma = 1$ ,*

| Symbol                                  | Meaning                                                             | Domain / locked value                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Rounds, arms, and context</b>        |                                                                     |                                                                                                                 |
| $t, T$                                  | Round index and horizon                                             | $t \in \{0, \dots, T-1\}; T = 10,000$                                                                           |
| $a, K$                                  | Provider arm and number of arms                                     | $K = 5$ : P-cheap, P-mid, P-premium, P-adv, P-flaky                                                             |
| $a_t$                                   | Arm selected at round $t$                                           | $a_t \in \mathcal{A}_t$                                                                                         |
| $\mathbf{x}_t$                          | Task context                                                        | Released encoder: $d = 7$ (task-type one-hot, difficulty, and two wallet features drawn from $H_t$ )            |
| $H_t$                                   | Pre-selection information available to the policy                   | Context, wallet state, affordable set, exposed provider metadata, and past purchase feedback                    |
| $k(\mathbf{x}_t), K_b$                  | Task-type bucket map used by PA-DCT                                 | T1/T2/T3a/T3b; $K_b = 4$                                                                                        |
| <b>Payment loop and feedback</b>        |                                                                     |                                                                                                                 |
| $\mathcal{Q}_t$                         | Quoted candidate set from discovery/pricing                         | Provider-price pairs before selection                                                                           |
| $p_{a,t}$                               | Provider charge at round $t$                                        | May drift; used for affordability and may be exposed as listed-price metadata                                   |
| $\mathcal{A}_t$                         | Affordable set, exposed as a mask before selection                  | $\mathcal{A}_t = \{a : p_{a,t} \leq B_t\}$                                                                      |
| $q_t$                                   | Observed quality of the chosen arm                                  | $[0, 1]$                                                                                                        |
| $f_t$                                   | Paid non-delivery flag (timeout, parse failure)                     | $\{0, 1\}$                                                                                                      |
| $c_t$                                   | Realized charge                                                     | $c_t = p_{a_t,t}$ , in USDC                                                                                     |
| $B, B_t$                                | Initial and remaining budget                                        | $B = \$50$ ; $B_{t+1} = B_t - c_t$                                                                              |
| $\tau$                                  | Completed purchases before termination or horizon end               | $\tau \leq T$                                                                                                   |
| <b>Reward and wallet pressure</b>       |                                                                     |                                                                                                                 |
| $\nu$                                   | Failure penalty in utility                                          | $\nu = 0.5$                                                                                                     |
| $u_t$                                   | Utility: failure-penalized quality; updates the Q-posterior         | $u_t = q_t - \nu f_t$                                                                                           |
| $c_{\max}, \tilde{c}_t$                 | Cost normalizer and normalized cost                                 | $c_{\max} = \$0.01$ ; $\tilde{c}_t = \text{clip}(c_t/c_{\max}, 0, 1)$                                           |
| $\text{burn\_dev}_t$                    | Deviation of the realized spend rate from the linear plan           | $\text{burn\_dev}_t = \frac{(B - B_t)/B}{t/T} - 1$ for $t > 0$ ; $\text{burn\_dev}_0 = 0$                       |
| $\alpha, \lambda_0$                     | Budget-pressure schedule constants                                  | $\alpha = 2.0, \lambda_0 = 1$                                                                                   |
| $\lambda_t, \lambda_{\text{norm}}$      | Raw and normalized wallet-pressure weights                          | $\lambda_t = \lambda_0 \exp(\alpha \text{burn\_dev}_t)$ ; $\lambda_{\text{norm}} = \lambda_t / (1 + \lambda_t)$ |
| $r_t$                                   | Payment-aware reward                                                | $r_t = (1 - \lambda_{\text{norm}})u_t - \lambda_{\text{norm}}\tilde{c}_t$                                       |
| <b>PA-DCT posteriors and selection</b>  |                                                                     |                                                                                                                 |
| $\bar{c}_a$                             | Initially listed cost of arm $a$                                    | Prior mean of the C-posterior                                                                                   |
| $\mu_{0,q}, \sigma_{0,q}^2, \sigma_q^2$ | Q-prior mean, prior variance, and likelihood variance               | 0.5, 1.0, 0.09                                                                                                  |
| $\sigma_{0,c}^2, \sigma_c^2$            | C-prior variance and likelihood variance                            | $10^{-4}, 10^{-6}$                                                                                              |
| $n_t^{a,k}, \bar{n}_t^{a,k}$            | Post-feedback and selection-time discounted counts of cell $(a, k)$ | Shared by the Q- and C-posteriors                                                                               |
| $S_t^{a,k}, S_{c_t}^{a,k}$              | Discounted utility and cost sums (barred: selection-time)           | New observations update only the chosen arm-bucket cell                                                         |
| $\gamma$                                | Common discount of the Q- and C-posteriors                          | 0.999 in main results                                                                                           |
| $\theta_q^{a,k}, \Sigma_q^{a,k}$        | Q-posterior mean and variance                                       | Normal-Normal posterior over utility                                                                            |
| $\theta_c^{a,k}, \Sigma_c^{a,k}$        | C-posterior mean and variance                                       | Normal-Normal posterior over realized cost                                                                      |
| $\hat{u}^{a,k}, \hat{c}^{a,k}$          | Thompson samples drawn at decision time                             | From the Q- and C-posteriors                                                                                    |
| $\hat{r}_a$                             | Decision-time sampled payment-aware reward                          | Mirrors $r_t$ with sampled utility and clipped sampled cost                                                     |
| <b>Reported evaluation quantities</b>   |                                                                     |                                                                                                                 |
| $\bar{q}_T$                             | Full-horizon mean quality                                           | Exhausted rounds count as $q_t = 0$                                                                             |
| $R_T^\pi, \text{PA-Reg}_T$              | Cumulative PA reward and empirical PA-regret                        | $\text{PA-Reg}_T = R_T^{\text{Oracle}} - R_T^\pi$ ; reported as $\text{PA-gap}/T$                               |
| ROI                                     | Quality per dollar                                                  | Efficiency diagnostic only                                                                                      |
| AdaptT                                  | Shock-response diagnostic in S2/S3                                  | Rounds to the first qualifying ROI window; defined in Appendix C                                                |

Table 5: Compact notation legend. Definitions remain in the main text; this table records conventions and the locked values used in all reported experiments.

define the policy-state PA-regret of clean-score PA-DCT against the wallet-coupled oracle as

$$g_t(a) := (1 - \lambda_{\text{norm}}) \mu_u^{a,k_t} - \lambda_{\text{norm}} \mu_c^{a,k_t} / c_{\max},$$

$$\text{PA-Reg}_T := \sum_{t=0}^{T-1} (g_t(a_t^*) - g_t(a_t)), \quad (17)$$

where  $a_t^* \in \arg \max_{a \in \mathcal{A}_t} g_t(a)$  at the policy's wallet pressure  $\lambda_{\text{norm}}$ , and  $a_t$  is PA-DCT's chosen arm. Conditional

on the bucket-visit sequence, clean-score PA-DCT reduces to per-bucket Gaussian Thompson sampling with a time-varying linear context. For a bucket visited  $T_k$  times,

$$\mathbb{E}[\text{PA-Reg}_{T_k}^{\text{cell } k}] \leq C_0 \sqrt{K T_k \log T_k}, \quad (18)$$

for a constant  $C_0$  depending on the prior and likelihood variances and on  $c_{\max}$ . Aggregating across  $K_b$  buckets with  $\sum_k T_k = T$  gives

$$\mathbb{E}[\text{PA-Reg}_T] \leq C \sqrt{K K_b T \log T}. \quad (19)$$

*Proof.* For a visit to bucket  $k$ , write  $w_t = \lambda_{\text{norm}}$  and define the observed feature

$$\phi_t = (1 - w_t, -w_t/c_{\max}), \quad \theta^{a,k} = (\mu_u^{a,k}, \mu_c^{a,k}).$$

Then  $g_t(a) = \phi_t^\top \theta^{a,k}$ . Under (A2'), the independent Q/C Thompson samples used by PA-DCT imply

$$(1 - w_t)\hat{u}^{a,k} - w_t\hat{c}^{a,k}/c_{\max} \sim \mathcal{N}(\phi_t^\top (\theta_q^{a,k}, \theta_c^{a,k}), (1 - w_t)^2 \Sigma_q^{a,k} + w_t^2 \Sigma_c^{a,k}/c_{\max}^2),$$

which is exactly the product-Gaussian posterior sample of  $\phi_t^\top \theta^{a,k}$ . This identifies the sampled score with a Gaussian posterior draw for the induced linear payoff.

The feature  $\phi_t$ , bucket  $k_t$ , and affordable set  $\mathcal{A}_t$  are observed before sampling. Bucket visits are determined by the exogenous task stream in §3, so conditioning on a bucket subsequence does not introduce policy-dependent context selection; the wallet pressure and affordable set are predictable state shared by the policy-state comparator in Eq. (17). Therefore the per-round gap is the standard contextual Thompson-sampling regret for at most  $K$  arms.

Consequently, on each bucket subsequence the clean-score policy is mathematically equivalent to Gaussian Thompson sampling on the induced linear payoff. Applying the Bayesian regret bound for  $K$ -armed Gaussian-prior Thompson sampling (Lattimore and Szepesvári 2020, Ch. 36) gives Eq. (18); the underlying confidence-bound decomposition of posterior sampling (Russo and Van Roy 2014, Prop. 1) carries over because  $\phi_t$  and  $\mathcal{A}_t$  are  $H_t$ -measurable, so the posterior-matching identity holds conditionally on  $H_t$ . Summing over buckets and using Cauchy–Schwarz,

$$\sum_k \sqrt{T_k \log T_k} \leq \sqrt{K_b T \log T}.$$

This gives Eq. (19).  $\square$

## B.6 Nonstationary analysis

We next isolate how exponential discounting removes stale evidence after an abrupt change. The result permits any fixed or reward-noise-independent revisit schedule, followed by a continuous-pull corollary with an explicit recovery horizon.

**Theorem 2** (Adaptation rate under an abrupt change, fixed schedule). *Fix  $\gamma \in (0, 1)$  and cell  $(a, k)$ . Suppose the true utility mean shifts between rounds  $t^* - 1$  and  $t^*$ , with pre-shock mean  $\mu_{\text{old}}$ , post-shock mean  $\mu_{\text{new}}$ , and  $\Delta = |\mu_{\text{old}} - \mu_{\text{new}}|$ . Consider any fixed, or reward-noise-independent, pull schedule for this cell covering pre- and post-shock pulls, with post-shock pull rounds  $\{s_1 < \dots < s_m\} \subseteq [t^*, t]$ . Define*

$$\begin{aligned} A &:= \gamma^{t-t^*+1} n_{t^*-1}^{a,k}, \\ B &:= \sum_{i=1}^m \gamma^{t-s_i}, \\ \kappa_q &:= \sigma_q^2 / \sigma_{0,q}^2. \end{aligned}$$

Here  $A$  is the discounted mass of stale pre-shock observations and  $B$  is the mass of new post-shock observations. Then the Bayesian  $Q$ -posterior mean satisfies

$$\left| \mathbb{E}[\theta_{q,t}^{a,k}] - \mu_{\text{new}} \right| \leq \frac{A\Delta + \kappa_q |\mu_{0,q} - \mu_{\text{new}}|}{\kappa_q + A + B}, \quad (20)$$

where the expectation is over observation noise for the fixed schedule.

**Remark 1** (Two regimes). The bound (20) separates a prior term from a stale-data term. When  $\kappa_q \gg A + B$ , the prior controls the bias. When  $\kappa_q \ll A + B$ , the leading term is  $\Delta A/(A + B)$ , which is reduced by discounting and by new post-shock pulls.

**Corollary 1** (Recovery horizon, data-dominated regime). *Suppose  $(a, k)$  is pulled at every post-shock round, so  $\ell := t - t^* + 1 = m$ , and assume the data-dominated regime  $\kappa_q \ll A + B$ . Then*

$$\left| \mathbb{E}[\theta_{q,t}^{a,k}] - \mu_{\text{new}} \right| \lesssim \Delta \gamma^\ell + O(\kappa_q/(A + B)),$$

and the leading stale-data term is at most  $\delta$  once

$$\ell \geq \frac{\log(\Delta/\delta)}{\log(1/\gamma)} = O(\log(\Delta/\delta)/(1 - \gamma)).$$

*Proof of Theorem 2.* Because the entire pull schedule is fixed or independent of the reward noise, conditioning on the pull times does not change the mean of the included observations; in particular, the pre-shock mass  $n_{t^*-1}^{a,k}$  is determined by the schedule. From Eq. (16),

$$\theta_{q,t}^{a,k} = \frac{\kappa_q \mu_{0,q} + S_t^{a,k}}{\kappa_q + n_t^{a,k}}.$$

Splitting the discounted mass into pre- and post-shock pulls gives  $n_t^{a,k} = A + B$  and  $\mathbb{E}[S_t^{a,k}] = A\mu_{\text{old}} + B\mu_{\text{new}}$ . Hence

$$\mathbb{E}[\theta_{q,t}^{a,k}] - \mu_{\text{new}} = \frac{\kappa_q(\mu_{0,q} - \mu_{\text{new}}) + A(\mu_{\text{old}} - \mu_{\text{new}})}{\kappa_q + A + B}.$$

Taking absolute values and using  $|\mu_{\text{old}} - \mu_{\text{new}}| = \Delta$  gives Eq. (20).  $\square$

*Proof of Corollary 1.* If the cell is pulled at every post-shock round, then  $B = \sum_{j=0}^{\ell-1} \gamma^j = (1 - \gamma^\ell)/(1 - \gamma)$  and  $A = \gamma^\ell n_{t^*-1}^{a,k}$ . By Lemma 1,  $n_{t^*-1}^{a,k} \leq 1/(1 - \gamma)$ , so

$$\frac{A}{A + B} \leq \frac{\gamma^\ell/(1 - \gamma)}{\gamma^\ell/(1 - \gamma) + (1 - \gamma^\ell)/(1 - \gamma)} = \gamma^\ell.$$

Substituting this into Eq. (20) yields the displayed bound; solving  $\Delta \gamma^\ell \leq \delta$  gives the stated horizon.  $\square$

**Remark 2** (Cost-posterior analogue). The same calculation applies to realized-cost shocks: replace  $(\mu_{\text{old}}, \mu_{\text{new}}, \mu_{0,q}, \kappa_q)$  with  $(\mu_{\text{old}}^c, \mu_{\text{new}}^c, \bar{c}_a, \kappa_c)$  and use the C-posterior statistic under the same locked discount  $\gamma$ . This is the algebra behind S3-style price changes once the changed arm is pulled again; we state the utility version to avoid duplicating notation.

## B.7 Scope of the formal analysis

The results above are component-level formal support, not a complete regret theorem for the full replay system. The stationary regret reduction assumes a well-specified product-Gaussian model,  $\gamma = 1$ , and the clean score with inactive

cost-sample clipping; the implemented clipped scorer coincides with this policy on trajectories where the clip is inactive. The adaptation result is conditional on a fixed, noise-independent pull schedule: it explains how discounted posteriors forget stale evidence after a changed arm is revisited, but it does not prove an unconditional Thompson-sampling revisit rate or a cumulative piecewise-stationary regret bound. Finally, the theorem’s comparator uses the policy’s own wallet pressure and affordable set, whereas the empirical *True Oracle* has its own wallet trajectory.

## C Detailed Ablation Study

This appendix reports the full metrics behind the component-role analysis in §6.4 of the main paper. The four primary variants each modify one component of Full PA-DCT:  $-P$  removes payment-aware ranking and scores providers by sampled utility alone;  $-D$  removes discounting ( $\gamma = 1$ );  $-C$  removes contextual bucketing by collapsing the four task-type buckets into one; and  $-TS$  replaces Thompson samples with posterior means. The additional price-shock diagnostic  $-C_{\text{post}}$  retains wallet pressure but pins the cost estimate to the specification cost  $\bar{c}_a$  and is evaluated only in S3. PA-gap/ $T$  and full-horizon mean quality are reported as endpoint metrics, whereas ROI and AdaptT serve as diagnostics. AdaptT is the number of rounds after the shock until the first trailing-200 ROI window reaches 95% of pre-shock ROI in S2 or 110% in S3. We average AdaptT over seeds that cross the threshold and report the corresponding hit count out of 30.

Two reading notes accompany Table 6. First, the  $-P$  row in S2 reports AdaptT over only the 14/30 seeds that reached the ROI threshold before exhausting the wallet. Its reported AdaptT is therefore conditional on the successful subset and should not be compared directly with the full-hit rows. Second,  $-P$  and Contextual DS-TS in Table 4 implement the same wallet-pressure-free selection rule through independent code paths and RNG streams. Their PA-gap/ $T$  values differ by at most 0.007 across scenarios, providing an implementation-level consistency check.

## D Sensitivity and Robustness Checks

This appendix reports the sensitivity checks behind the locked configuration in §6.1 of the main paper. We first sweep the four exposed selection scalars one at a time on S1, use the ablation panel for the cross-scenario discount check, and treat posterior variances,  $T$ , and  $B$  as fixed calibration constants rather than per-scenario tuning knobs. We then examine comparator-specific sweeps, the SW-TS diagnostic, and paired-seed statistical checks. Throughout, the goal is to verify that the locked configuration is a reasonable calibration, not to search for a better one. Comparator implementation details remain in Appendix F.

### D.1 Targeted S1 sensitivity sweep

Each non-default row in Table 7 changes one scalar while holding the others fixed, using 10 seeds at  $T = 10,000$ . The True Oracle is recomputed for each cell with the same  $\nu$  and

$c_{\text{max}}$  as the tested policy, so PA-gap/ $T$  remains comparable within the sweep.

The sweep supports two narrow claims. First, the locked policy is not brittle: no cell bankrupts the wallet,  $\bar{q}_T$  stays within  $[0.753, 0.802]$ , and PA-gap/ $T$  remains in the same order of magnitude. Second, the visible levers are the intended ones: shorter memory worsens stationary performance, and larger  $\alpha$  spends more aggressively;  $\nu$  and  $c_{\text{max}}$  have smaller effects in the tested range.

### D.2 Discount factor $\gamma$

The cross-scenario check uses the  $-D$  rows of Table 6: full PA-DCT sets  $\gamma = 0.999$ , while  $-D$  sets  $\gamma = 1$ . On S1, no discount is slightly better on the stationary endpoints, but discounting shortens AdaptT in both shock scenarios (1,467 vs. 2,249 rounds in S2; 200 vs. 279 in S3). This is the empirical counterpart of Cor. 1, where stale evidence decays as  $\gamma^\ell$  after a changed arm is revisited. The default  $\gamma = 0.999$  corresponds to an effective horizon of  $1/(1-\gamma) = 10^3$  rounds, chosen to retain S1 evidence while still reacting inside the S2 outage window.

### D.3 Other locked constants

The failure penalty  $\nu = 0.5$  gives billed timeouts a cost beyond their zero quality score; the sweep shows that halving or doubling it barely moves the metrics. Likewise, the pressure schedule uses  $\alpha = 2$  and  $\lambda_0 = 1$ , so  $\lambda_{\text{norm}} = 0.5$  when spending is on plan and rises smoothly under sustained overshoot; the sweep confirms the expected spend trade-off between  $\alpha = 1$  and  $\alpha = 4$ . The normalizer  $c_{\text{max}} = \$0.01$  is the highest listed provider price, and the half/double sweep does not change the qualitative conclusion.

The posterior prior and likelihood variances are intentionally not swept: the Q-posterior uses a broad mid-range prior with moderate proxy noise, while the C-posterior uses a tight prior and likelihood because realized cost is nearly deterministic except at the S3 price shock. We also do not sweep  $T$  or  $B$ : the released benchmark fixes  $T = 10,000$ , and  $B = \$50$  is calibrated so that a premium-only policy hits the budget cliff in the second half of the run.

### D.4 Comparator sensitivity sweeps

Table 8 sweeps the PM-Greedy threshold  $\theta_{\text{PM}}$  and the LinCBwK-Adapt exploration coefficient  $\beta$ ; the main table uses the locked settings  $\theta_{\text{PM}} = 0.7$  and  $\beta = 1.0$ . Reducing either  $\theta_{\text{PM}}$  or  $\beta$  improves the corresponding comparator’s PA-gap/ $T$  in every scenario. For reference, PA-DCT’s mean PA-gap/ $T$  values in the main results (Table 4 of the main paper) are 0.133, 0.166, and 0.121 in S1–S3. Across these comparator sweeps, PA-DCT retains the lowest mean PA-gap/ $T$  among the tested settings, with the margin largest in S1/S2 and narrower in S3.

### D.5 SW-TS diagnostic

SW-TS adapts sliding-window Thompson sampling (Trovò et al. 2020) to our admissible loop. It maintains a non-contextual Gaussian reward posterior per arm over utility observations from the last  $W = 1000$  global rounds and

| Scenario | Variant             | PA-gap/ $T$<br>↓ | ROI ↑ | Mean quality<br>↑  | AdaptT<br>(rounds) ↓ |
|----------|---------------------|------------------|-------|--------------------|----------------------|
| S1       | Full                | 0.133±0.004      | 378   | 0.797              | n/a                  |
| S1       | − $P$               | 0.855±0.058      | 111   | 0.555 <sup>‡</sup> | n/a                  |
| S1       | − $D$               | 0.109±0.008      | 412   | 0.810              | n/a                  |
| S1       | − $C$               | 0.116±0.003      | 390   | 0.812              | n/a                  |
| S1       | −TS                 | 0.130±0.046      | 535   | 0.740              | n/a                  |
| S2       | Full                | 0.166±0.006      | 357   | 0.761              | 1,467 (30/30)        |
| S2       | − $P$               | 0.878±0.050      | 103   | 0.517 <sup>‡</sup> | 1,611 (14/30)        |
| S2       | − $D$               | 0.170±0.012      | 399   | 0.745              | 2,249 (30/30)        |
| S2       | − $C$               | 0.158±0.006      | 368   | 0.761              | 1,176 (30/30)        |
| S2       | −TS                 | 0.175±0.048      | 537   | 0.687              | 1,125 (30/30)        |
| S3       | Full                | 0.121±0.004      | 429   | 0.831              | 200 (30/30)          |
| S3       | − $P$               | 0.259±0.036      | 351   | 0.840              | 200 (30/30)          |
| S3       | − $D$               | 0.114±0.009      | 426   | 0.838              | 279 (30/30)          |
| S3       | − $C$               | 0.109±0.003      | 425   | 0.848              | 398 (29/30)          |
| S3       | −TS                 | 0.145±0.025      | 552   | 0.742              | 2,232 (24/30)        |
| S3       | − $C_{\text{post}}$ | 0.144±0.005      | 423   | 0.797              | 200 (30/30)          |

Table 6: Full component-ablation metrics on 402Pilot-Bench. PA-gap/ $T$  is mean±SD over 30 paired seeds; ROI and quality entries are means. AdaptT is the mean elapsed rounds over hit seeds (hits/30), is lower-bounded by 200, and is a diagnostic rather than an endpoint. <sup>‡</sup> indicates that all 30 seeds exhausted the wallet before  $T$ ; n/a denotes a scenario without a shock.

| Setting                  | $\bar{q}_T$ | Budget % | ROI       | PA-gap/ $T$   |
|--------------------------|-------------|----------|-----------|---------------|
| DEFAULT                  | 0.796±0.001 | 42.2±0.3 | 377.5±2.1 | 0.1328±0.0013 |
| $\gamma = 0.99$          | 0.753±0.001 | 49.8±0.2 | 302.4±1.3 | 0.2087±0.0013 |
| $\gamma = 0.9995$        | 0.801±0.002 | 40.8±0.4 | 392.6±3.3 | 0.1218±0.0016 |
| $\nu = 0.25$             | 0.795±0.001 | 42.1±0.4 | 377.6±3.0 | 0.1329±0.0018 |
| $\nu = 1.0$              | 0.797±0.001 | 42.1±0.3 | 378.7±2.3 | 0.1324±0.0014 |
| $\alpha = 1.0$           | 0.788±0.002 | 36.7±0.3 | 430.3±3.1 | 0.1045±0.0011 |
| $\alpha = 4.0$           | 0.802±0.001 | 54.7±0.4 | 293.5±2.0 | 0.1781±0.0016 |
| $c_{\text{max}} = 0.005$ | 0.787±0.001 | 41.9±0.4 | 375.9±3.0 | 0.1567±0.0028 |
| $c_{\text{max}} = 0.02$  | 0.800±0.001 | 50.5±0.3 | 316.8±1.9 | 0.1490±0.0021 |

Table 7: One-at-a-time S1 sensitivity of PA-DCT. Means ± SE over 10 seeds; each non-default row perturbs a single scalar.

samples one utility draw per affordable arm. The window is indexed by global rounds: observations are timestamped by global round index and pruned once older than  $W$  rounds, whether or not the arm was recently pulled; if an arm’s active window is empty, the posterior reverts to the prior (mean 0.5, variance 1.0, likelihood variance 0.09). SW-TS does not learn cost and has no wallet-pressure term; the affordable-set mask is its only budget signal.

Table 9 reports the SW-TS endpoint metrics. For bankrupt seeds, aggregate metrics are computed over the full horizon, with post-bankruptcy rounds contributing zero quality and no further spend. Without wallet pressure, SW-TS repeatedly selects high-quality expensive arms; in S1/S2 this drains the wallet for all 30 seeds.

## D.6 Paired-seed statistical comparisons

For each scenario–metric pair, we compare PA-DCT with each baseline using the 30 within-seed differences. Reported  $p$ -values use two-sided paired normal-approximation tests. As a robustness check against skew from early budget exhaustion, we also compute 95% paired-bootstrap confidence

intervals with 10,000 resamples; these intervals support the same conclusions.

PA-DCT’s S3 improvements over Always-P-mid on quality, ROI, and PA-gap/ $T$  are significant (all  $p < 10^{-20}$ ), as is its S2 quality gain over Always-P-mid ( $p = 1.67 \times 10^{-3}$ ). Its S1 PA-gap/ $T$  disadvantage against Always-P-mid is also significant, consistent with exploration cost. In S2, PA-DCT’s PA-gap/ $T$  is not distinguishable from Always-P-cheap ( $p = 0.17$ ) or Contextual BTS ( $p = 0.86$ ). Against PM-Greedy, LinCBwK-Adapt, and Contextual DS-TS, its PA-gap/ $T$  advantage is significant in all three scenarios.

## E Allocation Dynamics under Market Shocks

Aggregate metrics summarize the overall quality–cost trade-off but do not show how the policy reallocates over time. Figure 2 reports PA-DCT’s provider-allocation trajectories across the three market scenarios. S1 serves as a stationary control; S2 shows withdrawal from and return to P-mid around the outage; and S3 shows migration toward discounted P-premium. These trajectories complement the AdaptT diagnostic in Table 6 by showing when and how

| Setting                                                         | Scenario | $\bar{q}_T$ | Budget % | PA-gap/ $T$ |
|-----------------------------------------------------------------|----------|-------------|----------|-------------|
| <i>PM-Greedy quality threshold <math>\theta_{PM}</math></i>     |          |             |          |             |
| $\theta_{PM} = 0.6$                                             | S1       | 0.776       | 55.3     | 0.244       |
| $\theta_{PM} = 0.6$                                             | S2       | 0.733       | 80.6     | 0.364       |
| $\theta_{PM} = 0.6$                                             | S3       | 0.793       | 35.4     | 0.152       |
| $\theta_{PM} = 0.7$                                             | S1       | 0.732       | 89.5     | 0.542       |
| $\theta_{PM} = 0.7$                                             | S2       | 0.627       | 98.7     | 0.660       |
| $\theta_{PM} = 0.7$                                             | S3       | 0.836       | 43.9     | 0.198       |
| $\theta_{PM} = 0.8$                                             | S1       | 0.667       | 94.6     | 0.684       |
| $\theta_{PM} = 0.8$                                             | S2       | 0.581       | 99.1     | 0.763       |
| $\theta_{PM} = 0.8$                                             | S3       | 0.836       | 46.8     | 0.243       |
| <i>LinCBwK-Adapt exploration coefficient <math>\beta</math></i> |          |             |          |             |
| $\beta = 0.5$                                                   | S1       | 0.803       | 70.0     | 0.312       |
| $\beta = 0.5$                                                   | S2       | 0.773       | 83.5     | 0.386       |
| $\beta = 0.5$                                                   | S3       | 0.806       | 37.8     | 0.164       |
| $\beta = 1.0$                                                   | S1       | 0.824       | 93.0     | 0.457       |
| $\beta = 1.0$                                                   | S2       | 0.796       | 99.9     | 0.515       |
| $\beta = 1.0$                                                   | S3       | 0.836       | 44.5     | 0.195       |
| $\beta = 2.0$                                                   | S1       | 0.832       | 100.0    | 0.515       |
| $\beta = 2.0$                                                   | S2       | 0.800       | 100.0    | 0.533       |
| $\beta = 2.0$                                                   | S3       | 0.854       | 45.7     | 0.197       |

Table 8: One-at-a-time sensitivity sweeps for PM-Greedy and LinCBwK-Adapt. Means over 30 paired seeds.

| Scenario | $\bar{q}_T$ | Budget % | ROI | PA-gap/ $T$ | Bankrupt |
|----------|-------------|----------|-----|-------------|----------|
| S1       | 0.483       | 100.0    | 97  | 0.981       | 30/30    |
| S2       | 0.466       | 100.0    | 93  | 0.986       | 30/30    |
| S3       | 0.855       | 52.3     | 328 | 0.324       | 0/30     |

Table 9: SW-TS appendix results. Means over 30 paired seeds; bankruptcy is the number of seeds that exhausted the wallet before  $T = 10,000$ .

provider allocation changes.

## F Extended Comparators

### F.1 Comparator Implementations

This section describes how each comparator is implemented in *402Pilot-Bench* under the common frozen-replay evaluation protocol. Where necessary, standard algorithms are minimally adapted to operate under replay while preserving their original decision principles. All non-oracle implementations receive the same task context and hard affordable-set mask. After each decision, the loop reveals only the chosen provider’s failure-penalized utility and realized receipt cost; each implementation uses only the signals specified below. No comparator observes counterfactual cached outcomes, and no learning comparator uses a forced round-robin warm start.

**Random and fixed-arm policies.** We implement Random by sampling uniformly from the affordable set. Always-P-cheap, Always-P-mid, and Always-P-premium return their named provider when it is affordable and otherwise return the first provider in the runtime’s fixed affordable-set ordering. These policies ignore context and replay feedback.

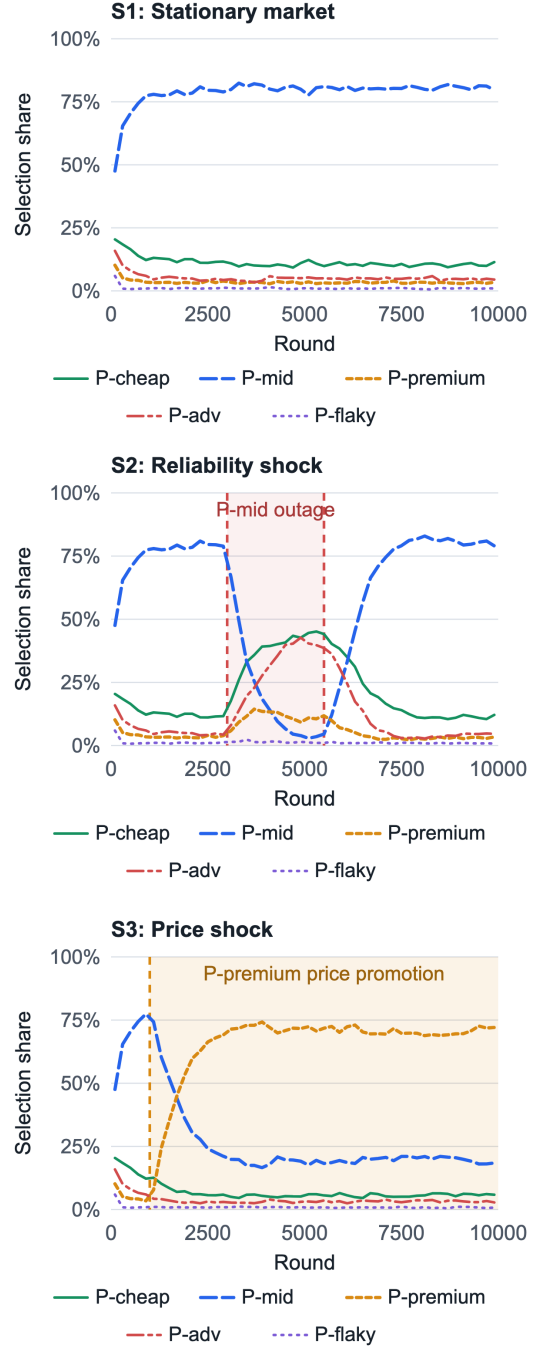


Figure 2: PA-DCT provider-allocation shares over  $T = 10,000$  rounds in, from top to bottom, the stationary market (S1), reliability-shock scenario (S2), and price-shock scenario (S3). In S1, P-mid remains dominant after initial exploration. In S2, the policy shifts away from P-mid during its outage and returns after recovery. In S3, it reallocates toward discounted P-premium.

**BudgetRule.** We implement BudgetRule as a deterministic wallet-threshold policy. It selects P-premium when the

remaining-budget fraction is above 0.50, P-mid when it lies in  $(0.20, 0.50]$ , and P-cheap otherwise. If the target tier is unavailable, it selects the affordable provider with the highest catalog price. It ignores context and replay feedback.

**Contextual DS-TS.** In our implementation, Contextual DS-TS adapts discounted Thompson sampling (Qi, Wang, and Zhu 2023) to four task buckets and continuous utility feedback. It maintains one Gaussian utility posterior per provider–task-bucket cell, discounts every cell’s sufficient statistics once per replay round, and samples one utility value for each affordable provider in the current bucket. Only the chosen cell is updated. Receipt cost and wallet state do not enter the rule beyond the runtime’s affordable-set mask.

**Contextual BTS.** In our replay implementation, Contextual BTS adapts budgeted Thompson sampling (Xia et al. 2015) to four task buckets and continuous utility and cost feedback. It maintains stationary Gaussian utility and cost posteriors for each provider–task-bucket cell, with each cost prior centered at that provider’s catalog price. For every affordable provider, it samples both quantities and selects the largest sampled utility-to-cost ratio; the sampled cost is floored at  $10^{-6}$  USDC. Only the chosen cell is updated from its utility and receipt cost, and wallet state enters only through affordability.

**PM-Greedy.** PM-Greedy is implemented as a price-metadata router inspired by cheapest-sufficient routing (Chen, Zaharia, and Zou 2024). For each provider–task-bucket cell, it keeps a per-pull window of the 500 most recent observed utilities; a cell with no observations yet uses the optimistic estimate 1. Among affordable providers whose mean utility is at least 0.7, it selects the one with the lowest scenario-aware posted price, breaking price ties by higher estimated utility. If none clears the threshold, it selects the highest-utility provider, breaking ties by lower posted price. Receipt cost is not used for learning, and the affordable-set mask is the only wallet signal.

**LinCBwK-Adapt.** Our replay adaptation of LinCBwK (Agrawal and Devanur 2016) is an admissible empirical comparator rather than a reproduction of the full theoretical algorithm. With the four-dimensional one-hot task context, its linear reward and cost models reduce to stationary Gaussian/ridge estimates for each provider–task-bucket cell. It normalizes receipt cost by  $c_{\max} = \$0.01$  and scores affordable providers by  $UCB_r - \mu LCB_c$ . Before each selection,  $\mu$  is increased by 0.01 when the remaining-budget fraction is below the remaining-horizon fraction and decreased by 0.01 otherwise, subject to the range  $[0, 100]$ . Only the chosen cell’s utility and normalized receipt cost are updated; no temporal forgetting is applied.

**Replay True Oracle.** We implement the replay True Oracle with an independent wallet trajectory. For the same replay task and cached-response version, it inspects the realized outcome of every provider affordable under that wallet, evaluates each using the oracle wallet’s current payment pressure, and selects the provider with the highest payment-aware reward. Only the selected receipt is charged. This counterfactual ac-

cess makes the oracle unattainable and suitable only as an upper bound.

## F.2 Comparator Configurations

Table 10 reports the fixed configurations of comparators with explicit policy parameters, including the appendix-only SW-TS diagnostic of Appendix D.5.

**Alignment and numerical guards.** Contextual DS-TS shares PA-DCT’s Q-prior, likelihood variance, and discount. Contextual BTS shares PA-DCT’s Q/C prior settings and likelihood variances but removes discounting, reducing differences due solely to prior calibration. LinCBwK-Adapt instead uses narrower ridge-style prior variances and does not seed its normalized-cost prior from catalog prices. The  $10^{-6}$  cost floors in Contextual BTS and LinCBwK-Adapt prevent sampled or lower-confidence costs near zero from destabilizing their selection scores.

## G x402 Integration Witness

The released artifact includes a local x402 integration witness for the `PaymentExecutor` boundary described in §4 of the main paper. The witness demonstrates that a selected-provider purchase can execute as a real HTTP 402 quote–pay–receipt loop. It is separate from 402Pilot-Bench and contributes no reported measurements. The local setup consists of an Anvil fork containing USDC contract state, a Python x402 facilitator for verifying and settling exact EVM payments, a FastAPI resource server exposing x402-protected provider endpoints, and an `X402PaymentExecutor` adapter that maps the HTTP 402 exchange to a 402Pilot Outcome.

**Runtime environment.** The reported benchmark experiments are deterministic, CPU-only frozen-replay runs conducted on a MacBook Pro with an Apple M2 Max and 32 GB unified memory. The released artifact pins the software dependencies and exact versions. The integration witness additionally requires a local Anvil instance, the Python x402 facilitator, and the FastAPI resource server. Running `bash scripts/witness_smoke.sh` exercises both the unpaid HTTP 402 response and a paid request, and verifies a valid payment receipt together with the corresponding decrease in the buyer’s local Anvil USDC balance.

| Comparator       | Configuration                                                                                                                                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BudgetRule       | High and low remaining-budget thresholds are 0.50 and 0.20, respectively.                                                                                                                                                                |
| Contextual DS-TS | $K_b = 4$ ; Q-prior mean, prior variance, and likelihood variance (0.5, 1.0, 0.09); wall-clock discount $\gamma = 0.999$ .                                                                                                               |
| Contextual BTS   | $K_b = 4$ ; Q settings (0.5, 1.0, 0.09); C-prior mean $\bar{c}_a$ ; C-prior and likelihood variances ( $10^{-4}$ , $10^{-6}$ ); $\gamma = 1$ ; sampled-cost floor $10^{-6}$ USDC.                                                        |
| PM-Greedy        | $K_b = 4$ ; per-pull window $W = 500$ ; utility threshold $\theta_{PM} = 0.7$ ; optimistic initial estimate $\hat{u}_{init} = 1$ .                                                                                                       |
| LinCBwK-Adapt    | $K_b = 4$ ; reward and normalized-cost prior means 0.5; corresponding prior and likelihood variances 0.09; $\gamma = 1$ ; $c_{max} = \$0.01$ ; $\beta = 1.0$ ; dual step $\eta = 0.01$ and range $[0, 100]$ ; cost-LCB floor $10^{-6}$ . |
| SW-TS diagnostic | Non-contextual; wall-clock window $W = 1000$ ; reward-prior mean and variance (0.5, 1.0); likelihood variance 0.09.                                                                                                                      |

Table 10: Comparator configurations used in the reported experiments. Parameters are fixed across scenarios and random seeds unless explicitly varied in Appendix D.

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